

# The Foundations: Logic and Proofs

## Chapter 1, Part II: Predicate Logic

# Propositional Logic Summary

## The Language of Propositions

- Connectives: Negation  $\neg$  Conjunction  $\wedge$  Disjunction  $\vee$   
Implication  $\rightarrow$  Biconditional  $\leftrightarrow$  Exclusive Or  $\oplus$
- Truth Values
- Truth Tables

## Logical Equivalences

- Equivalent Propositions

# Propositional Equivalences Summary

1, Tautologies 恒真式, Contradictions 矛盾式, and Contingencies 适然式

2, Logically Equivalent

Two compound propositions  $p$  and  $q$  are **logically equivalent** if  $p \leftrightarrow q$  is **a tautology**.

Two compound propositions  $p$  and  $q$  are **equivalent** if and only if the columns in a **truth table** giving their truth values agree.

We can show that **two expressions are logically equivalent by developing a series of logically equivalent statements**.

*conditional-disjunction equivalence:*  $\neg p \vee q \equiv p \rightarrow q$

De Morgan's Laws  $\neg(p \wedge q) \equiv \neg p \vee \neg q$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

3, Key Logical Equivalences

# Predicates and Quantifiers (谓词和量词)

## Section 1.4

## 1.4.2 Predicates

Example: The statement “ $x$  is greater than 0” has two parts. The first part, the variable  $x$ . The second part, the **predicate**, “is greater than 0”.

We can denote the statement “ $x$  is greater than 0” by  $P(x)$ ,  $P$  denotes the **predicate** “is greater than 0” and  $x$  is the **variable**.

The statement  $P(x)$  is also said to be the value of the **propositional function**(命题函数)  $P$  at  $x$ .

Once a value has been assigned to the variable  $x$ , the statement  $P(x)$  becomes a proposition and has truth value.

For example, let  $P(x)$  denote “ $x > 0$ ” and the domain be the integers. Then:

$P(-3)$  is false.

$P(3)$  is true.

Often the **domain** is denoted by  $U$ . So in this example  $U$  is the integers.

# Introducing Predicate(谓词) Logic

**Predicate logic** uses the following new features:

- Variables:  $x, y, z$
- Predicates:  $P(x), M(x)$
- Quantifiers (*to be covered in a few slides*):

***Propositional functions*** are a generalization of propositions.

- They contain variables and a predicate, e.g.,  $P(x)$
- Variables can be replaced by elements from their *domain*.

# Examples of Propositional Functions

Let “ $x + y = z$ ” be denoted by  $R(x, y, z)$  and  $U$  (for all three variables) be the integers. Find these truth values:

$R(2, -1, 5)$

**Solution:** F

$R(3, 4, 7)$

**Solution:** T

$R(x, 3, z)$

**Solution:** Not a Proposition

## Compound Expressions

Connectives from propositional logic carry over to predicate logic.

If  $P(x)$  denotes “ $x > 0$ ,” find these truth values:

$P(3) \vee P(-1)$

**Solution:** T

$P(3) \wedge P(-1)$

**Solution:** F

$P(3) \rightarrow P(-1)$

**Solution:** F

$P(3) \rightarrow \neg P(-1)$

**Solution:** T

# 1.4.3 Quantifiers



Charles  
Peirce  
(1839-1914)

We need *quantifiers*(量词) to express the meaning of English words including *all* and *some*:

- “All men are Mortal.”
- “Some cats do not have fur.”

The two most important quantifiers are:

- *Universal Quantifier*, (全称量词) “For all,” symbol:  $\forall$  (任意)
- *Existential Quantifier*, (存在量词) “There exists,” symbol:  $\exists$  (存在)

We write as in  $\forall x P(x)$  and  $\exists x P(x)$ .

$\forall x P(x)$  asserts  $P(x)$  is true for every  $x$  in the *domain*.

$\exists x P(x)$  asserts  $P(x)$  is true for some  $x$  in the *domain*.

**TABLE 1** Quantifiers.

| <i>Statement</i> | <i>When True?</i>                         | <i>When False?</i>                         |
|------------------|-------------------------------------------|--------------------------------------------|
| $\forall x P(x)$ | $P(x)$ is true for every $x$ .            | There is an $x$ for which $P(x)$ is false. |
| $\exists x P(x)$ | There is an $x$ for which $P(x)$ is true. | $P(x)$ is false for every $x$ .            |



# Universal Quantifier

$\forall x P(x)$  is read as “For all  $x$ ,  $P(x)$ ” or “For every  $x$ ,  $P(x)$ ”

## Examples:

- 1) If  $P(x)$  denotes “ $x > 0$ ” and  $U$  is the integers, then  $\forall x P(x)$  is false.
- 2) If  $P(x)$  denotes “ $x > 0$ ” and  $U$  is the positive integers, then  $\forall x P(x)$  is true.
- 3) If  $P(x)$  denotes “ $x$  is even” and  $U$  is the integers, then  $\forall x P(x)$  is false.

The meaning of the **universal quantification** of  $P(x)$  changes when we change the **domain**. The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not defined.

# Existential Quantifier

$\exists x P(x)$  is read as “For some  $x$ ,  $P(x)$ ”, or as “There is an  $x$  such that  $P(x)$ ,” or “There is at least one  $x$  such that  $P(x)$ .”

## Examples:

1. If  $P(x)$  denotes “ $x > 0$ ” and  $U$  is the integers, then  $\exists x P(x)$  is true. It is also true if  $U$  is the positive integers.
2. If  $P(x)$  denotes “ $x < 0$ ” and  $U$  is the positive integers, then  $\exists x P(x)$  is false .
3. If  $P(x)$  denotes “ $x$  is even” and  $U$  is the integers, then  $\exists x P(x)$  is true.

# Properties of Quantifiers

The **truth value** of  $\exists x P(x)$  and  $\forall x P(x)$  **depend on both the propositional function  $P(x)$  and on the domain  $U$ .**

## Examples:

1. If  $U$  is the positive integers and  $P(x)$  is the statement “ $x < 2$ ”, then  $\exists x P(x)$  is true, but  $\forall x P(x)$  is false.

# 1.4.4 Thinking about Quantifiers as Conjunctions and Disjunctions

If the domain is finite, a universally quantified proposition is equivalent to a conjunction of propositions without quantifiers and an existentially quantified proposition is equivalent to a disjunction of propositions without quantifiers.

If  $U$  consists of the integers 1,2, and 3:

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$$

Even if the domains are infinite, you can still think of the quantifiers in this fashion, but the equivalent expressions without quantifiers will be infinitely long.

## 1.4.6 Precedence of Quantifiers

The quantifiers  $\forall$  and  $\exists$  have higher precedence than all the logical operators.

For example,  $\forall x P(x) \vee Q(x)$  means  $(\forall x P(x)) \vee Q(x)$ ,

$\forall x (P(x) \vee Q(x))$  means something different.

## 1.4.8 Equivalences in Predicate Logic

Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value

- for every predicate substituted into these statements and
- for every domain of discourse 话语 used for the variables in the expressions.

The notation  $S \equiv T$  indicates that  $S$  and  $T$  are logically equivalent.

**Example:**  $\forall x \neg\neg S(x) \equiv \forall x S(x)$

# 1.4.9 Negating Quantified Expressions<sub>1</sub>

Consider  $\forall x J(x)$

“Every student in your class has taken a course in Java.”

Here  $J(x)$  is “ $x$  has taken a course in Java” and

the domain is students in your class.

Negating the original statement gives “It is not the case that every student in your class has taken Java.” This implies that “There is a student in your class who has not taken Java.”

Symbolically  $\neg \forall x J(x)$  and  $\exists x \neg J(x)$  are equivalent

# Negating Quantified Expressions<sub>2</sub>

Now Consider  $\exists x J(x)$

“There is a student in this class who has taken a course in Java.”

Where  $J(x)$  is “ $x$  has taken a course in Java,” the domain is students in your class.

Negating the original statement gives “It is not the case that there is a student in this class who has taken Java.” This implies that “Every student in this class has not taken Java”

Symbolically  $\neg \exists x J(x)$  and  $\forall x \neg J(x)$  are equivalent



# De Morgan's Laws for Quantifiers

The rules for negating quantifiers are:

| TABLE 2 De Morgan's Laws for Quantifiers. |                             |                                            |                                        |
|-------------------------------------------|-----------------------------|--------------------------------------------|----------------------------------------|
| <i>Negation</i>                           | <i>Equivalent Statement</i> | <i>When Is Negation True?</i>              | <i>When False?</i>                     |
| $\neg \exists x P(x)$                     | $\forall x \neg P(x)$       | For every $x$ , $P(x)$ is false.           | There is $x$ for which $P(x)$ is true. |
| $\neg \forall x P(x)$                     | $\exists x \neg P(x)$       | There is an $x$ for which $P(x)$ is false. | $P(x)$ is true for every $x$ .         |

The reasoning in the table shows that:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

These are important.

# 1.4 Predicates and Quantifiers Section Summary

Predicates  $P$

Variables  $x$

Propositional function  $P(x)$

Quantifiers

- Universal Quantifier  $\forall$
- Existential Quantifier  $\exists$

Negating Quantifiers

- De Morgan's Laws for Quantifiers

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

# Exercise

**10.** Let  $C(x)$  be the statement “ $x$  has a cat,” let  $D(x)$  be the statement “ $x$  has a dog,” and let  $F(x)$  be the statement “ $x$  has a ferret.” Express each of these statements in terms of  $C(x)$ ,  $D(x)$ ,  $F(x)$ , quantifiers, and logical connectives. Let the domain consist of all students in your class.

- a)** A student in your class has a cat, a dog, and a ferret.
- b)** All students in your class have a cat, a dog, or a ferret.
- c)** Some student in your class has a cat and a ferret, but not a dog.
- d)** No student in your class has a cat, a dog, and a ferret.

**10. a)** We assume that this means that one student has all three animals:  $\exists x(C(x) \wedge D(x) \wedge F(x))$ .

**b)**  $\forall x(C(x) \vee D(x) \vee F(x))$       **c)**  $\exists x(C(x) \wedge F(x) \wedge \neg D(x))$

**d)** This is the negation of part (a):  $\neg \exists x(C(x) \wedge D(x) \wedge F(x))$ .

# Exercise

**11.** Let  $P(x)$  be the statement “ $x = x^2$ .” If the domain consists of the integers, what are these truth values?

**b)**  $P(1)$

**c)**  $P(2)$

**e)**  $\exists xP(x)$

**f)**  $\forall xP(x)$

**19.** Suppose that the domain of the propositional function  $P(x)$  consists of the integers 1, 2, 3, 4, and 5. Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.

**a)**  $\exists xP(x)$

**d)**  $\neg\forall xP(x)$

# Nested Quantifiers

## Section 1.5

# 1.5.1 Nested Quantifiers

**Nested quantifiers**(嵌套量词) are often necessary to express the meaning of sentences in English as well as important concepts in computer science and mathematics.

**Example:** “Every real number has an inverse” is

$$\forall x \exists y (x + y = 0)$$

where the **domains** of  $x$  and  $y$  are the real numbers.

We can also think of nested propositional functions:

$\forall x \exists y (x + y = 0)$  can be viewed as  $\forall x Q(x)$  where  $Q(x)$  is  $\exists y P(x, y)$  where  $P(x, y)$  is  $(x + y = 0)$

## 1.5.3 Order of Quantifiers

### Examples:

1. Let  $P(x,y)$  be the statement “ $x + y = y + x$ .” Assume that  $U$  is the real numbers. Then  $\forall x \forall y P(x,y)$  and  $\forall y \forall x P(x,y)$  have the same truth value.
2. Let  $Q(x,y)$  be the statement “ $x + y = 0$ .” Assume that  $U$  is the real numbers. Then  $\forall x \exists y Q(x,y)$  is true, but  $\exists y \forall x Q(x,y)$  is false.

# Questions on Order of Quantifiers<sub>1</sub>

**Example 1:** Let  $U$  be the real numbers,

Define  $P(x,y) : x \cdot y = 0$

What is the truth value of the following:

1,  $\forall x \forall y P(x, y)$  False

2,  $\forall x \exists y P(x, y)$  True

3,  $\exists x \forall y P(x, y)$  True

4,  $\exists x \exists y P(x, y)$  True



# Quantifications of Two Variables

| Statement                                                      | When True?                                                  | When False                                                   |
|----------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------|
| $\forall x \forall y P(x, y)$<br>$\forall y \forall x P(x, y)$ | $P(x, y)$ is true for every pair $x, y$ .                   | There is a pair $x, y$ for which $P(x, y)$ is false.         |
| $\forall x \exists y P(x, y)$                                  | For every $x$ there is a $y$ for which $P(x, y)$ is true.   | There is an $x$ such that $P(x, y)$ is false for every $y$ . |
| $\exists x \forall y P(x, y)$                                  | There is an $x$ for which $P(x, y)$ is true for every $y$ . | For every $x$ there is a $y$ for which $P(x, y)$ is false.   |
| $\exists x \exists y P(x, y)$<br>$\exists y \exists x P(x, y)$ | There is a pair $x, y$ for which $P(x, y)$ is true.         | $P(x, y)$ is false for every pair $x, y$                     |

# 1.5.4 Translating Mathematical Statements into Predicate Logic

**Example :** Translate “The sum of two positive integers is always positive” into a logical expression.

**Solution:**

1. Rewrite the statement to make the implied quantifiers and domains explicit:  
“For every two integers, if these integers are both positive, then the sum of these integers is positive.”
2. Introduce the variables  $x$  and  $y$ , and specify the domain, to obtain:  
“For all positive integers  $x$  and  $y$ ,  $x + y$  is positive.”
3. The result is:

$$\forall x \forall y ((x > 0) \wedge (y > 0)) \rightarrow (x + y > 0)$$

where the **domain** of both variables consists of **all integers**

We can express this as:  $\forall x \forall y (x + y > 0)$ , where the **domain** for both variables consists of **all positive integers**.

# Section Summary<sub>2</sub>

Nested Quantifiers

Order of Quantifiers

Translating Mathematical Statements into  
Statements involving Nested Quantifiers.

# The Foundations: Logic and Proofs

## Chapter 1, Part III: Proofs

# Rules of Inference

## Section 1.6

## 1.6.2 Valid Arguments<sub>1</sub>

We will show how to construct **valid arguments** (有效论证) in two stages; first for **propositional logic** and then for **quantified statements**. **The rules of inference** (推理规则) are the essential building block in the construction of **valid arguments**.

1. Propositional Logic
2. Quantified Statements

## 1.6.3 Rules of Inference for Propositional Logic: Modus Ponens(离断律,肯定前件)

$$p \rightarrow q$$

**Corresponding Tautology:**

$$\frac{p}{\therefore q}$$

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

**Example:**

Let  $p$  be “It is snowing.”

Let  $q$  be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“It is snowing.”

“Therefore , I will study discrete math.”

## 1.6.3 Rules of Inference for Propositional Logic: Modus Tollens (否定后件)

$$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$$

**Corresponding Tautology:**

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

**Example:**

Let  $p$  be “it is snowing.”

Let  $q$  be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“I will not study discrete math.”

“Therefore , it is not snowing.”



# Rules of Inference for Propositional Logic:

## Hypothetical Syllogism(假设三段论)

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

**Corresponding Tautology:**

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

**Example:**

Let  $p$  be “it snows.”

Let  $q$  be “I will study discrete math.”

Let  $r$  be “I will get an A.”

“If it snows, then I will study discrete math.”

“If I study discrete math, I will get an A.”

“Therefore , If it snows, I will get an A.”

# Disjunctive Syllogism(分取三段论)

$$\begin{array}{l} p \vee q \\ \hline \neg p \\ \hline \therefore q \end{array}$$

**Corresponding Tautology:**

$$(\neg p \wedge (p \vee q)) \rightarrow q$$

**Example:**

Let  $p$  be “I will study discrete math.”

Let  $q$  be “I will study English literature.”

“I will study discrete math or I will study English literature.”

“I will not study discrete math.”

“Therefore , I will study English literature.”

# Addition

**Corresponding Tautology:**

$$\frac{p}{\therefore p \vee q}$$

$$p \rightarrow (p \vee q)$$

# Simplification(简单化)

$$\frac{p \wedge q}{\therefore p}$$

**Corresponding Tautology:**

$$(p \wedge q) \rightarrow p$$

# Conjunction(合取式,与)

$$\frac{p}{\frac{q}{\therefore p \wedge q}}$$

**Corresponding Tautology:**

$$((p) \wedge (q)) \rightarrow (p \wedge q)$$

# Resolution(消解算法)

$$\frac{p \vee q}{\frac{\neg p \vee r}{\therefore q \vee r}}$$

**Corresponding Tautology:**

$$((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$$

**TABLE 1** Rules of Inference.

| <i>Rule of Inference</i>                                                                               | <i>Tautology</i>                                                             | <i>Name</i>            |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------|
| $\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$                             | $(p \wedge (p \rightarrow q)) \rightarrow q$                                 | Modus ponens           |
| $\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$                   | $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$                       | Modus tollens          |
| $\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$ | $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ | Hypothetical syllogism |
| $\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$                               | $((p \vee q) \wedge \neg p) \rightarrow q$                                   | Disjunctive syllogism  |
| $\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$                                         | $p \rightarrow (p \vee q)$                                                   | Addition               |
| $\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$                                       | $(p \wedge q) \rightarrow p$                                                 | Simplification         |
| $\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$                                  | $((p) \wedge (q)) \rightarrow (p \wedge q)$                                  | Conjunction            |
| $\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$                 | $((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$                 | Resolution             |

## 1.6.4 Using the Rules of Inference(推理规则) to Build Valid Arguments(有效论证)

**Example 1:** From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that  $q$  is a conclusion.

**Solution:**

| Step                            | Reason                         |
|---------------------------------|--------------------------------|
| 1. $p \wedge (p \rightarrow q)$ | Premise                        |
| 2. $p$                          | Simplification using (1)       |
| 3. $p \rightarrow q$            | Simplification using (1)       |
| 4. $q$                          | Modus Ponens using (2) and (3) |

## Example 2: Valid Arguments<sub>3</sub>

With these hypotheses:

“It is not sunny this afternoon and it is colder than yesterday.”

“We will go swimming only if it is sunny.”

“If we do not go swimming, then we will take a canoe trip.”

“If we take a canoe trip, then we will be home by sunset.”

Using the inference rules, construct a valid argument for the conclusion:

“We will be home by sunset.”

### Solution:

1. Choose propositional variables:

$p$  : “It is sunny this afternoon.”    $q$  : “It is colder than yesterday.”

$r$  : “We will go swimming.”    $s$  : “We will take a canoe trip.”

$t$  : “We will be home by sunset.”

Translation into propositional logic:

Hypotheses:  $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion:  $t$

*Continued on next slide →*

# Valid Arguments<sub>4</sub>

Hypotheses:  $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion:  $t$

## 3. Construct the Valid Argument

| Step                      | Reason                          |
|---------------------------|---------------------------------|
| 1. $\neg p \wedge q$      | Premise                         |
| 2. $\neg p$               | Simplification using (1)        |
| 3. $r \rightarrow p$      | Premise                         |
| 4. $\neg r$               | Modus tollens using (2) and (3) |
| 5. $\neg r \rightarrow s$ | Premise                         |
| 6. $s$                    | Modus ponens using (4) and (5)  |
| 7. $s \rightarrow t$      | Premise                         |
| 8. $t$                    | Modus ponens using (6) and (7)  |



# 1.6.7 Rules of inference for quantified statements

## Universal Instantiation (UI)(全称实例)

$$\frac{\forall xP(x)}{\therefore P(c)}$$

**Example:**

Our domain consists of all dogs and Fido is a dog.

“All dogs are cuddly(可爱的).”

“Therefore, Fido is cuddly.”

## Universal Generalization (UG)(全称推广)

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall xP(x)}$$

Used often implicitly in Mathematical Proofs.

# Existential Instantiation (EI)(存在实例)

$$\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$$

**Example:**

“There is someone who got an A in the course.”

“Let’s call her *Amy* and say that *Amy* got an A”

# Existential Generalization (EG)(存在推广)

$$\frac{P(c) \text{ for some element } c}{\therefore \exists xP(x)}$$

**Example:**

“Michelle got an A in the class.”

“Therefore, someone got an A in the class.”

**TABLE 2** Rules of Inference for Quantified Statements.

| <i>Rule of Inference</i>                                            | <i>Name</i>                |
|---------------------------------------------------------------------|----------------------------|
| $\frac{\forall xP(x)}{\therefore P(c)}$                             | Universal instantiation    |
| $\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall xP(x)}$ | Universal generalization   |
| $\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$ | Existential instantiation  |
| $\frac{P(c) \text{ for some element } c}{\therefore \exists xP(x)}$ | Existential generalization |

# Using Rules of Inference<sub>1</sub>

**Example 1:** Using the rules of inference, construct a valid argument to show that

“John Smith has two legs”

is a consequence of the premises:

“Every man has two legs.” “John Smith is a man.”

**Solution:** Let  $M(x)$  denote “ $x$  is a man” and  $L(x)$  “ $x$  has two legs” and let John Smith be a member of the domain.

**Valid Argument:**

| Step                                  | Reason                         |
|---------------------------------------|--------------------------------|
| 1. $\forall x(M(x) \rightarrow L(x))$ | Premise                        |
| 2. $M(J) \rightarrow L(J)$            | UI from (1)                    |
| 3. $M(J)$                             | Premise                        |
| 4. $L(J)$                             | Modus Ponens using (2) and (3) |

# Using Rules of Inference<sub>2</sub>

**Example 2:** Use the rules of inference to construct a valid argument showing that the conclusion

“Someone who passed the first exam has not read the book.”

follows from the premises

“A student in this class has not read the book.”

“Everyone in this class passed the first exam.”

**Solution:** Let  $C(x)$  denote “ $x$  is in this class,”  $B(x)$  denote “ $x$  has read the book,” and  $P(x)$  denote “ $x$  passed the first exam.”

First we translate the premises and conclusion into symbolic form(象征形式).

$$\begin{array}{c} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \\ \hline \therefore \exists x(P(x) \wedge \neg B(x)) \end{array}$$

*Continued on next slide →*

# Using Rules of Inference<sub>3</sub>

## Valid Argument:

| Step                                  | Reason                  |
|---------------------------------------|-------------------------|
| 1. $\exists x(C(x) \wedge \neg B(x))$ | Premise                 |
| 2. $C(a) \wedge \neg B(a)$            | EI from (1)             |
| 3. $C(a)$                             | Simplification from (2) |
| 4. $\forall x(C(x) \rightarrow P(x))$ | Premise                 |
| 5. $C(a) \rightarrow P(a)$            | UI from (4)             |
| 6. $P(a)$                             | MP from (3) and (5)     |
| 7. $\neg B(a)$                        | Simplification from (2) |
| 8. $P(a) \wedge \neg B(a)$            | Conj from (6) and (7)   |
| 9. $\exists x(P(x) \wedge \neg B(x))$ | EG from (8)             |

# Using Rules of Inference<sub>2</sub>

**Example 2:** Use the rules of inference to construct a valid argument showing that the conclusion

“Someone who passed the first exam has not read the book.”

follows from the premises

“A student in this class has not read the book.”

“Everyone in this class passed the first exam.”

**Solution:** Let  $C(x)$  denote “ $x$  is in this class,”  $B(x)$  denote “ $x$  has read the book,” and  $P(x)$  denote “ $x$  passed the first exam.”

First we translate the premises and conclusion into symbolic form(象征形式).

$$\frac{\begin{array}{l} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x(P(x) \wedge \neg B(x))}$$

Why not  $\exists x(C(x) \rightarrow \neg B(x))$ ? If there is an  $x$ ,  $x \notin C(x)$ , then  $C(x) \rightarrow \neg B(x)$  is true. This is not the condition we want to talk about.

Why not  $\forall x(C(x) \wedge P(x))$ ? This means, for all people, they are in this class and have passed the exam, this is also not the premise talk about.

*Continued on next slide →*

Conclusion:

for universal quantifier, we use implication “ $\rightarrow$ ”

For existential quantifier, we use conjunction “ $\wedge$ ”



# 1.6.8 Combining rules of inference for propositions and quantified statements

Universal Modus Ponens(全称肯定前件) combines universal instantiation and modus ponens into one rule.

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ P(a), \text{ where } a \text{ is a particular} \\ \text{element in the domain} \\ \hline \therefore Q(a) \end{array}$$

Universal Modus Tollens (全称否定后件)

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ \neg Q(a), \text{ where } a \text{ is a particular element in the domain} \\ \hline \therefore \neg P(a) \end{array}$$

# Section Summary<sub>1</sub>

Valid Arguments

Inference Rules for Propositional Logic

Using Rules of Inference to Build Arguments

Rules of Inference for Quantified Statements

**TABLE 1** Rules of Inference.

| <i>Rule of Inference</i>                                                                               | <i>Tautology</i>                                                             | <i>Name</i>            |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------|
| $\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$                             | $(p \wedge (p \rightarrow q)) \rightarrow q$                                 | Modus ponens           |
| $\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$                   | $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$                       | Modus tollens          |
| $\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$ | $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ | Hypothetical syllogism |
| $\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$                               | $((p \vee q) \wedge \neg p) \rightarrow q$                                   | Disjunctive syllogism  |
| $\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$                                         | $p \rightarrow (p \vee q)$                                                   | Addition               |
| $\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$                                       | $(p \wedge q) \rightarrow p$                                                 | Simplification         |
| $\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$                                  | $((p) \wedge (q)) \rightarrow (p \wedge q)$                                  | Conjunction            |
| $\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$                 | $((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$                 | Resolution             |

**TABLE 2** Rules of Inference for Quantified Statements.

| <i>Rule of Inference</i>                                            | <i>Name</i>                |
|---------------------------------------------------------------------|----------------------------|
| $\frac{\forall xP(x)}{\therefore P(c)}$                             | Universal instantiation    |
| $\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall xP(x)}$ | Universal generalization   |
| $\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$ | Existential instantiation  |
| $\frac{P(c) \text{ for some element } c}{\therefore \exists xP(x)}$ | Existential generalization |

# Homework

Page 57: 14 a c, 35 a c.

Page 70: 19 a b, 28 a b.

Page 82: 5, 14 a b.

**14.** Determine the truth value of each of these statements if the domain consists of all real numbers.

**b)**  $\exists x(x^4 < x^2)$

**d)**  $\forall x(2x > x)$

**35.** Express the negation of each of these statements in terms of quantifiers without using the negation symbol.

**b)**  $\forall x(x \leq 2)$

**d)**  $\exists x(x < 0)$

- 19.** Express each of these statements using mathematical and logical operators, predicates, and quantifiers, where the domain consists of all integers.
- c)** The sum of the squares of two integers is greater than or equal to the square of their sum.
  - d)** The absolute value of the product of two integers is the product of their absolute values.

**28.** Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.

**c)**  $\exists x \forall y (xy = 0)$

**d)**  $\exists x \exists y (x + y \neq y + x)$

6. Use rules of inference to show that the hypotheses “If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on,” “If the sailing race is held, then the trophy will be awarded,” and “The trophy was not awarded” imply the conclusion “It rained.”

6. Let  $r$  be the proposition “It rains,” let  $f$  be the proposition “It is foggy,” let  $s$  be the proposition “The sailing race will be held,” let  $l$  be the proposition “The life saving demonstration will go on,” and let  $t$  be the proposition “The trophy will be awarded.” We are given premises  $(\neg r \vee \neg f) \rightarrow (s \wedge l)$ ,  $s \rightarrow t$ , and  $\neg t$ . We want to conclude  $r$ . We set up the proof in two columns, with reasons, as in Example 6. Note that it is valid to replace subexpressions by other expressions logically equivalent to them.

| Step                                                         | Reason                              |
|--------------------------------------------------------------|-------------------------------------|
| 1. $\neg t$                                                  | Hypothesis                          |
| 2. $s \rightarrow t$                                         | Hypothesis                          |
| 3. $\neg s$                                                  | Modus tollens using (1) and (2)     |
| 4. $(\neg r \vee \neg f) \rightarrow (s \wedge l)$           | Hypothesis                          |
| 5. $(\neg(s \wedge l)) \rightarrow \neg(\neg r \vee \neg f)$ | Contrapositive of (4)               |
| 6. $(\neg s \vee \neg l) \rightarrow (r \wedge f)$           | De Morgan’s law and double negative |
| 7. $\neg s \vee \neg l$                                      | Addition, using (3)                 |
| 8. $r \wedge f$                                              | Modus ponens using (6) and (7)      |
| 9. $r$                                                       | Simplification using (8)            |



**14.** For each of these arguments, explain which rules of inference are used for each step.

c) “All movies produced by John Sayles are wonderful. John Sayles produced a movie about coal miners. Therefore, there is a wonderful movie about coal miners.”

c) Let  $s(x)$  be “ $x$  is a movie produced by Sayles,” let  $c(x)$  be “ $x$  is a movie about coal miners,” and let  $w(x)$  be “movie  $x$  is wonderful.” We are given premises  $\forall x(s(x) \rightarrow w(x))$  and  $\exists x(s(x) \wedge c(x))$ , and we want to conclude  $\exists x(c(x) \wedge w(x))$ . In our proof,  $y$  represents an unspecified particular movie.

| Step                                  | Reason                               |
|---------------------------------------|--------------------------------------|
| 1. $\exists x(s(x) \wedge c(x))$      | Hypothesis                           |
| 2. $s(y) \wedge c(y)$                 | Existential instantiation using (1)  |
| 3. $s(y)$                             | Simplification using (2)             |
| 4. $\forall x(s(x) \rightarrow w(x))$ | Hypothesis                           |
| 5. $s(y) \rightarrow w(y)$            | Universal instantiation using (4)    |
| 6. $w(y)$                             | Modus ponens using (3) and (5)       |
| 7. $c(y)$                             | Simplification using (2)             |
| 8. $w(y) \wedge c(y)$                 | Conjunction using (6) and (7)        |
| 9. $\exists x(c(x) \wedge w(x))$      | Existential generalization using (8) |